REPORT

L.D. COLLEGE OF ENGINEERING, AHMEDABAD



SMART TRAFFIC SIGNAL CONTROLLER

Subject: Fuzzy logic & Genetic Programming

Submitted By:

Name: Patel Dip N. (240280106096)

Patel Vraj M.(240280106113)

Patel Aastha M.(240280106092)

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1. PROBLEM STATEMENT

1.1 Introduction

Traffic congestion is a major problem in urban areas due to the increasing number of vehicles on roads. Conventional traffic signals operate using fixed time intervals, regardless of the actual traffic density at an intersection.

For example, a road with a large number of vehicles may receive insufficient green time, while a road with very few vehicles may receive the same green time. This results in unnecessary waiting, fuel consumption, air pollution, and traffic congestion.

A Smart Traffic Signal Controller using Fuzzy Logic is designed to overcome these problems by adjusting signal timings according to real-time traffic conditions.

1.2 Problem Definition

The main problem is the inefficient allocation of green signal time at road intersections. A fixed-time traffic system cannot respond effectively to changing traffic density.

Therefore, an intelligent traffic control system is required to estimate traffic demand and allocate suitable green time to different approaches.

2. OBJECTIVES

The main objectives of the project are:

1. To develop an intelligent traffic signal control system using fuzzy logic.
2. To reduce vehicle waiting time at road intersections.
3. To minimize traffic congestion by adjusting green signal duration.
4. To use traffic density as an input for signal timing decisions.
5. To provide a flexible and adaptive traffic control mechanism.
6. To reduce unnecessary vehicle stops, fuel consumption, and emissions.
7. To improve the efficiency of traffic movement at intersections.
8. To demonstrate the application of fuzzy logic in intelligent transportation systems.

3. SYSTEM DESIGN

3.1 Overview of the System

The proposed system uses fuzzy logic to determine the green signal duration based on traffic conditions.

Traffic density is measured using sensors, cameras, or simulated vehicle-count data. The collected information is processed by a fuzzy logic controller, which determines an appropriate green time for the selected road.

The controller uses linguistic variables such as Low, Medium, and High to represent traffic density and applies predefined fuzzy rules to determine the output.

3.2 Components of the System

1. Traffic Sensors

Sensors or cameras are used to detect vehicles and estimate traffic density on each road.

2. Input Processing Unit

This unit receives the traffic data and converts it into a suitable format for the fuzzy logic controller.

3. Fuzzy Logic Controller

The controller processes traffic density using membership functions and fuzzy IF–THEN rules.

4. Signal Timing Unit

This unit converts the controller output into a suitable green signal duration.

5. Traffic Signal Lights

Red, yellow, and green lights control vehicle movement at the intersection.

6. Display and Monitoring Unit

This unit can display the estimated traffic density and the selected signal timing.

3.3 Block Diagram

Traffic Sensors / Camera

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Traffic Density Measurement

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Fuzzification

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Fuzzy Rule Evaluation

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Defuzzification

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Green Time Calculation

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Traffic Signal Control

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Monitoring and Feedback

3.4 Working Principle

The system first measures the number of vehicles waiting on each road. The measured traffic density is converted into fuzzy values such as Low, Medium, and High.

The fuzzy inference system evaluates the input values using predefined rules. The resulting fuzzy output is converted into a numerical green time through defuzzification.

The signal controller then assigns an appropriate green duration to the selected road, subject to minimum and maximum timing limits and safety constraints.

4. FUZZY LOGIC AND MEMBERSHIP FUNCTIONS

4.1 Introduction to Fuzzy Logic

Fuzzy logic is a method of decision-making that handles uncertain or imprecise information. Unlike conventional binary logic, fuzzy logic allows a value to belong partially to more than one category.

For example, traffic density may be classified as 60% Medium and 40% High rather than being assigned to only one category.

This feature makes fuzzy logic suitable for traffic signal control because traffic conditions change continuously.

4.2 Input Variable

Traffic Density

Traffic density is represented by the number of vehicles waiting on a road approach.

For this illustrative model, the input range is assumed to be 0–40 vehicles.

The fuzzy sets are:

- Low Traffic: 0–15 vehicles
- Medium Traffic: 10–30 vehicles
- High Traffic: 25–40 vehicles

These ranges overlap to allow smooth transitions between categories. They are illustrative design values and should be calibrated using actual traffic data.

4.3 Output Variable

Green Signal Time

The output of the fuzzy logic controller is the green signal duration.

For this illustrative model, the output range is 10–60 seconds.

The fuzzy sets are:

- Short Green Time: 10–25 seconds
- Medium Green Time: 20–45 seconds
- Long Green Time: 40–60 seconds

Higher traffic demand generally results in a longer green duration, subject to safety limits and the requirements of other approaches.

4.4 Types of Membership Functions

The following membership functions can be used:

1. Triangular Membership Function:
It uses three parameters and represents a fuzzy set with a triangular shape.

2. Trapezoidal Membership Function:
It uses four parameters and represents a fuzzy set with a trapezoidal shape.

3. Gaussian Membership Function:
It uses a smooth bell-shaped curve and provides gradual transitions between fuzzy sets.

For a simple traffic signal controller, triangular and trapezoidal membership functions are easy to implement and explain.

5. FUZZY RULES

5.1 Introduction

Fuzzy rules are IF–THEN statements used to determine the output according to traffic conditions. The rules represent the decision-making logic of the traffic signal controller.

5.2 Rule Base

The following eight rules can be used in the proposed system.

Rule 1: IF traffic density is Low, THEN green signal time is Short.

Rule 2: IF traffic density is Medium, THEN green signal time is Medium.

Rule 3: IF traffic density is High, THEN green signal time is Long.

Rule 4: IF traffic density is Very High, THEN green signal time is Very Long, subject to the maximum permitted green time.

Rule 5: IF traffic density is Low and the waiting time is Short, THEN green signal time is Short.

Rule 6: IF traffic density is Medium and the waiting time is Long, THEN green signal time is Medium to Long.

Rule 7: IF traffic density is High and the waiting time is Long, THEN green signal time is Long.

Rule 8: IF traffic density is Low but the waiting time is Very Long, THEN the controller should provide a suitable service opportunity to prevent excessive delay.

The final green time is selected after evaluating the applicable rules and combining their outputs.

Note: Rules involving waiting time require waiting time to be included as an additional input variable. If the controller uses traffic density as its only input, implement Rules 1–4 and define additional density-only rules as required.

5.3 Rule Evaluation

The controller evaluates the degree to which each rule applies. The outputs of the active rules are combined to produce a final fuzzy output.

A defuzzification method, such as the centroid method, converts the combined fuzzy output into a numerical green time.

6. IMPLEMENTATION

6.1 Implementation Method

The proposed controller can be implemented using MATLAB Fuzzy Logic Toolbox, Python, or a microcontroller-based system.The basic implementation steps are:

1. Collect or simulate traffic data for each road approach.
2. Define the input and output variables.
3. Create membership functions for traffic density and green signal time.
4. Define the fuzzy IF–THEN rules.
5. Select a fuzzy inference method, such as Mamdani inference.
6. Apply a defuzzification method, such as the centroid method.
7. Calculate the green signal duration.
8. Apply minimum green, maximum green, yellow, and all-red safety intervals as required.
9. Monitor traffic conditions and repeat the process.

6.2 Algorithm

Step 1: Start the system.

Step 2: Read traffic density from the sensors or simulated data.

Step 3: Convert the input into fuzzy membership values.

Step 4: Evaluate the fuzzy rule base.

Step 5: Combine the rule outputs.

Step 6: Defuzzify the output to obtain the green signal duration.

Step 7: Check the duration against permitted timing limits.

Step 8: Operate the signal sequence safely.

Step 9: Update traffic measurements and repeat the process.

6.3 Pseudocode

START

Read traffic density.

Fuzzify traffic density.

Evaluate fuzzy rules.

Combine the fuzzy outputs.

Calculate green time using defuzzification.

Limit green time to the permitted range.

Apply the safe signal transition sequence.

Update traffic measurements.

Repeat the process.

STOP

6.4 Safety Considerations

The fuzzy logic controller must not directly switch conflicting traffic movements to green simultaneously.

The implementation should include:

- Minimum and maximum green times.
- Yellow signal intervals.
- All-red clearance intervals where required.
- Conflict-free phase sequencing.
- Pedestrian crossing requirements.
- Emergency vehicle priority where supported.
- Maximum waiting-time limits to prevent starvation of low-volume approaches.

The controller should follow applicable traffic signal standards and local authority requirements.

7. RESULTS

7.1 Expected Results

The fuzzy logic-based traffic signal controller is expected to provide more flexible signal timing than a fixed-time controller when traffic demand varies significantly.

The expected outcomes are:

1. Longer green times for approaches with higher traffic demand.
2. Shorter green times for approaches with lower traffic demand, when appropriate.
3. Better adaptation to changing traffic conditions.
4. Potential reduction in average vehicle waiting time.
5. Potential reduction in queue length and unnecessary vehicle stops.
6. Improved utilization of available green time.

7.2 Illustrative Output Table

The following values demonstrate the intended controller behaviour.

Traffic Density	Example Green Time	Signal Decision
Low (5 vehicles)	15 seconds	Short green
Medium (20 vehicles)	30 seconds	Medium green
High (35 vehicles)	50 seconds	Long green

They are example values, not results from an actual simulation or field test.

The actual output depends on the membership functions, rule base, defuzzification method, and timing constraints.

7.3 Comparison with Fixed-Time Control

Parameter	Fixed-Time Controller	Fuzzy Logic Controller
Signal timing	Predetermined	Adapted to measured demand
Traffic response	Limited	Flexible
High traffic demand	May receive insufficient time	Can receive longer green time
Low traffic demand	May receive unnecessary green time	Can reduce allocated green time
Implementation	Relatively simple	Requires sensors and controller logic
Performance verification	Timing-based evaluation	Requires simulation or field testing

8. ANALYSIS AND DISCUSSION

8.1 Performance Analysis

The performance of the proposed controller should be evaluated using traffic simulation or real-world measurements.

Important performance indicators include:

1. Average Waiting Time:
The average time vehicles spend waiting at the intersection.

2. Average Queue Length:
The average number of vehicles waiting on each road approach.

3. Vehicle Throughput:
The number of vehicles passing through the intersection per unit time.

4. Number of Stops:
The number of times vehicles are required to stop during their journey through the intersection.

5. Fuel Consumption and Emissions:
These can be estimated using a suitable traffic or emissions model.

The fuzzy logic controller should be compared with a fixed-time controller under the same traffic demand, road geometry, and simulation conditions

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8.2 Advantages

1. Adapts signal timing to changing traffic conditions.
2. Can reduce unnecessary waiting under variable traffic demand.
3. Uses understandable IF–THEN decision rules.
4. Supports multiple traffic density categories.
5. Can be integrated with sensors, cameras, and intelligent transportation systems.
6. Can be expanded to include waiting time, pedestrian demand, and emergency vehicle priority.

8.3 Limitations

1. Accurate traffic measurement is required.
2. Sensors and cameras may be affected by weather or technical faults.
3. Membership functions and rules require proper calibration.
4. A poorly designed rule base may increase delays on some approaches.
5. Installation and maintenance may increase system cost.
6. The expected improvement must be verified through simulation or field testing.

8.4 Future Scope

The system can be improved by incorporating:

- Artificial intelligence and machine learning.
- Real-time camera-based vehicle detection.
- Emergency vehicle detection and priority.
- Pedestrian demand estimation.
- Coordination between nearby intersections.
- Adaptive control using live traffic data.
- Integration with intelligent transportation systems.

9. CONCLUSION

The Smart Traffic Signal Controller using Fuzzy Logic provides an intelligent approach to traffic signal timing. Unlike a conventional fixed-time system, the proposed controller uses traffic conditions to determine a suitable green signal duration.

The system includes traffic data collection, fuzzification, rule evaluation, defuzzification, and safe signal operation. It can potentially reduce waiting time and congestion when its rules are properly designed and calibrated.

However, the actual improvement in traffic performance cannot be confirmed without simulation or field testing. Future development can incorporate additional inputs, such as waiting time, pedestrian demand, and emergency vehicle priority, to improve the system further.

10. REFERENCES

1. L. A. Zadeh, “Fuzzy Sets,” Information and Control, Vol. 8, No. 3, pp. 338–353, 1965.
2. Timothy J. Ross, Fuzzy Logic with Engineering Applications, Wiley.
3. MathWorks, MATLAB Fuzzy Logic Toolbox Documentation.
4. Transportation Research Board, Highway Capacity Manual.

Relevant traffic signal design guidelines and standards issued by the applicable road and traffic authorities.